

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2013

Programme: BE

Full Marks: 100

Course: Programming Technology

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain Document-View architecture with its advantages. 7
b) Introduce C# programming with the help of its basic data types, literals, constants, operators and variables. 8
- OR**
- How namespace system works in C# programming. Write a C# program to demonstrate the concept of object and class.
2. a) What is MVC architecture? Explain the pros and cons of 2-tier, 3-tier and N-tier programming architecture. 7
b) Explain about NET framework with its advantages. How JIT helps to improve the runtime performance of computer program? 8
3. a) What is server control in asp.net? Explain asp.net page life cycle. 7
b) What is the difference between error and exception? Write a sample program which illustrates "try", "catch" and "finally" for exception handling in Java. 8
4. a) Verify the statement "JVM is a platform independent engine". 5
b) Explain JSP core tags, formatting tags, XML tags and SQL tags with suitable examples. Illustrate. 5
c) Differentiate JDBC with ODBC. 5
5. a) How does applet differ from applications? Explain the purposes of different attributes of Applet tags. 7
b) What is delegation event model? Is it possible to process events without using the delegation event model? Explain with an example. 8

OR

Differentiate between AWT and Swing. How do we handle mouse events in Java?

Approximation ADC with the reference voltage of 10 volts.

- b) What are the various types of data transmission? Explain Synchronous & Asynchronous Data transmission in brief.
- 6. a) Explain the principle of operation of strip chart recorder.

Or

Explain the operation of X-Y recorder with its application.

- b) What are the uses of data acquisition system?
- 7. Write short notes on: (Any two)
 - a) Errors in instrumentation
 - b) Type of transducers
 - c) Filtering

Level: Bachelor
Programme: BE
Course: Programming Technology

Semester: Spring

Year : 2014
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

- a) Differentiate between object oriented and event oriented programming. 8
- b) What are the advantages of document view architecture? Explain in brief about the components of visual programming. 7
- a) What is MVC architecture? How MVC processes the request and response. 8
- b) How does C# support encapsulating the method. Explain with an example. 7
- a) What is JIT (Just in Time) compilation? How does it work in Dot Net Framework? Explain with a block diagram. 7
- b) What is Dynamic Web programming? Make comparison between ASP and ASP.NET. 8
- a) Write a java program to read and write object stream. 7
- b) Differentiate between java application and java applet. Write a java applet to display an image. 8
- a) How is event handling done by swing? Write a java swing program with a button. Display "Hello Java" when the button is clicked. 8
- b) What is JSP and Servlet technology? Explain in brief about the advantages of Servlet over the CGI technology. 7
- a) What are the interfaces of HTTP request handling? Mention each with their name and short description. Explain in brief about servlet life cycle. 7
- b) Differentiate between JDBC and ODBC. For the table given below, write SQL statements for performing the following task. 8

- i. Create the table "Student".
- ii. Insert one row with values for each field.
- iii. Display name, address and DOB for the person whose DOB is greater than 2000.

ID	Name	Address	Gender	DOB
1	Ram	Pokhara	Male	2002
2	Gita	Kathmandu	Female	1995

7. Write short notes on: **(Any two)**

2×5

- a) JVM and JRE
- b) WPF
- c) Data source object.

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Attempt all the questions.

- a) Explain about the significance of **aspect orient** and **subject oriented programming** with example. 7
- b) What is client server model? Describe the relationship between Model, View and Controller in MVC architecture. 8
- a) What are the C# collection classes? Define the term: Class, Object, Event and delegate. 8
- b) What is windows form? Differentiate Windows communication foundation (WCF) with Windows presentation Foundation (WPF). 7
- a) What is ASP? Illustrate a simple database program using ASP. 8
- b) What do you mean by **user defined exceptions**? Write a Java program to read a text file and display in console. 7
- a) What are the parameters to applets? Also describe different applet security policies. 7
- b) How does **Java event handling** work? Write a swing application with one close button when clicked, it should terminate the application. 8
- a) How does JSP differ with Java Servlet? Explain the life cycle of JSP. 8
- b) Illustrate with code, how JDBC is used to connect the program with the database. 7
- a) Give the HTML to create a form with two elements: a textbox named *First Name* that holds a maximum of 50 characters, and a *Submit* button. The form should submit its data to a JSP program called *Process Name.jsp* using the POST method. Also write the *Process name.jsp* to handle these requests. 7
- b) What are the types of **JDBC statements** available? Explain with example. 8

Write short notes on: (Any two)

2×5

- a) Components of Visual Programming
- b) CLS
- c) JVM and JRE.

Level: Bachelor
Programme: BE
Course: Programming Technology

Semester: Fall

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

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Attempt all the questions.

- a) What do you mean by IDE? What are the difference between events oriented and object oriented programming. 8
- b) Write merits and demerits of MVC? Why is MVC a three tier architecture? 7
2. a) Write a program in C# to input the names of n cities and display them in alphabetical order. 7
- b) What do you mean by Versioning, Namespace and Delegates? Write a code to calculate factorial using recursive function. 8
3. a) What is ASP? Write a program to display data from database with ASP. 8
- b) What is a java class file? How does java attain platform independence? 7
4. a) Differentiate between throw and throws with suitable example. 8
- b) What are Applet security policies as compared to application? Give the suitable example to access parameter value to applet application. 7
5. a) What is event delegation model? Write a program with a button. Change the background color to green on clicking the button. 8
- b) Write a code for following example using Servlet. 7

Username:	
Password:	
submit reset	

6. a) What are the uses of **session**? How is it managed in **JSP**? 8
- b) How can you access data from database server to your application using JAVA where database name is db_Inventory and table is tbl_Customer? 7
7. Write short notes on: (**Any two**) 2×5
- a) Action Listener.
- b) ASP.net
- c) Window object.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Programming Technology

Semester: Spring

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe the features of aspect oriented programming and subject oriented programming? 7
b) Explain document/view architecture? What are four key classes used in this architecture? 8
2. a) Write merits and demerits of MVC? What is the difference between 2-tier and 3-tier architecture. 7
b) What are keywords and identifiers? Explain different datatypes used in C#. 8
3. a) Discuss the merits of JIT compilation. Also, explain the process of garbage collection. 7
b) A database dbBook has a table called tbBookInfo having attributes bookid, bookname, authorname, publisher and price. Using ADO.net and ASP.net write a program to read all the records from the table and display them. 8
4. a) Briefly explain the JVM, JDK and JRE. 7
b) Explain different streams used in java. Write a java program to read a file called "Employee.txt" and display its contents it in console. 8
5. a) How can you convert a java application into java applet? Explain with an example code for both applet and application. 8
b) What is event delegation model? Write a code for mouse event delegation model with your own example. 7
6. a) What are the uses of session? How is it managed in JSP? Explain with examples. 8
b) What is JDBC? Explain different drivers used in JDBC. 7

7. Write short notes on: (**Any two**)

- a) JSP
- b) Building GUI with swing
- c) Steps for connecting database in JDBC
- d) Servlet Life Cycle

Level: Bachelor
Programme: BE
Course: Programming Technology

Semester: Fall

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe the features of Procedure, Event and Object programming? 7
b) What do you mean by Thick and Thin Client? Explain N-tier Architecture. 8
2. a) What is delegate in C#? Illustrate with a meaningful example. 7
b) How dot net application is created? Explain each process in brief. 8
3. a) How to make database connectivity in asp.net? Explain with example. 5
b) Explain, why Java is called a platform independent language? 5
c) Define Exception in Java. How it is handled? Explain with example. 5
4. a) Write a swing application with two input textboxes and a button which when clicked, it should write the result of two string in textboxes to console after converting them to Uppercase. 8
b) Explain about the lifecycle of Applet? Write a simple but meaning applet program. 7
5. a) Explain HTTP requests handling and Session management in JSP. 8
b) Explain about the life cycle of JSP with a neat diagram? 7
6. a) Define Server side database. Explain the types of Server side database drives and tools. 8
b) Write a JSP application that submits the basic student registration form to another page called procoessRegister.jsp which should display all the records. 7
7. Write short notes on: (Any two) 2×5
 - a) Document View Architecture
 - b) Event handling
 - c) CLR and CTS

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Programming Technology

Semester: Spring

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define event oriented and aspect oriented programming. Explain virtual machine and run time environments. 8
b) What is MVC architecture? Explain pros and cons of 2-tier and 3-tier architecture. 7
2. a) Define Delegate. Differentiate between simple array and ArrayList? Explain with example. 7
b) Explain .NET framework with suitable diagram. How JIT (Just in Time) compilation helps to improve the runtime performance of computer program? 8
3. a) What is the ASP.NET? Describe about ASP.NET page life cycle. 7
b) Verify the statement "JVM is a platform independent engine". 8
4. a) What is Exception? How can you handle Exception in Java? Explain with example. 7
b) What is applet? Explain Applet architecture and security. 8
5. a) What is event and Event delegation model? 7
b) Give the HTML to create a form with two elements: a textbox named *First Name* that holds a maximum of 50 characters, and a *Submit* button. The form should submit its data to a JSP program called *Process Name.jsp* using the POST method. Also write the *Process name.jsp* to handle these requests. 8
6. a) Explain Servlet life cycle and How can you deploy Servlet in Java. 7
b) What are the types of **JDBC statements** available? Explain with example. 8
7. Write short notes on: (Any two)
a) Document view architecture
b) Namespace in C#
c) Swing 2×5

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2017

Programme: BE

Full Marks: 100

Course: Programming Technology

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe the features of four different language paradigms and give one example language for each category. 7
b) What is an IDE and how does it work? Discuss the key features in Visual Studio .NET. 8
2. a) What is difference between 3-layer architecture and MVC architecture? 8
b) Describe the steps involved in establishing database connectivity in asp.net with code snippets? 7
3. a) Describe some notable distinguishing features of C#. Discuss relationship between C# and .NET Framework. 8
b) What is jagged array in C#.Net? What is delegates in C# and uses of delegates? 7
4. a) Discuss the architecture of Java virtual machine and its component using suitable diagrams. 8
b) Define Error and Exception in Java. What is the difference between throw, throws and Throwable in java? 7
5. a) What is applet? Explain the lifecycle of an applet and demonstrate the same using a simple program. 8
b) Discuss these basic components of Swing with code snippets – JButton, JTextField, JCheckBox, JRadioButton. 7
6. a) What are the session management techniques available JSP? Explain briefly. 7
b) How to Connect to The Database in Java? Discuss with code snippets. 8
7. Write short notes on: (Any two) 8
a) CLR and CTS 2×5
b) Event Handling in Java
c) Servlet Life Cycle

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) What is IDE? What are the components of Visual Programming?
b) Differentiate between:
 - iii. Thin and Thick Clients
 - iv. 2 Tier, 3 Tier and n-Tier Architecture
2. a) What are arrays in strings? How are they implemented in C#? Illustrate with an example.
b) What are WCF and WPF? Describe about the different base classes used in .NET Framework.
3. a) Write a program to create a class Student. This class must possess general attributes of the students as well as marks in 5 subjects. Illustrate different methods to take input and show the output, the different attributes of the class.
b) Illustrate with a small example, how mouse event handling is done using Java?
c) What is JRE? Why do we need JRE? Explain in brief.
4. a) What is exception handling? Illustrate with an example how exceptions are handled in a Java Program.
b) Write a code to create a simple applet in Java. Describe the difference between applet and application.
5. a) What is swing? How GUI is built using swing?
b) Define JSP and Servlets. How are servlets created and deployed? Explain with an illustration.
6. a) What package is used to connect database with .NET? Explain the syntax of the connection mechanism.
b) Illustrate with an example, how database is connected using JDBC?
7. Write short notes on: **(Any two)**
 - a) Document View Architecture and its advantages
 - b) Garbage Collection
 - c) Session Management
 - d) Server side database drivers

Level: Bachelor
Programme: BE
Course: Programming Technology

Semester: Fall

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define aspect oriented programming. What are the difference between procedures oriented and object oriented programming? 8
b) Illustrate MVC with diagram? What are the differences between 2-tier and 3-tier architecture of software development. 7
2. a) Define namespace and versioning in C#. How does C# solve real world problem? 7
b) Explain about Windows Presentation Foundation and Common Type System. What are the advantages of Dot Net Framework? 8
3. a) Discuss the steps involved in connecting to a SQL Database using ASP.NET. 8
b) What is checked and unchecked exception in Java? Write a program to create your own exception class in java. 7
4. a) Explain Applet Architecture. How can you convert applet to application? Explain with suitable example. 8
b) Write differences between AWT and Swing? Write a simple program for handling Event in java. 7
5. a) Explain JSP and Servlet. What are the advantages of JSP? 7
b) Define JavaServer Pages Standard Tag Library (JSTL) and HttpSession Interface. Explain JSP life cycle 8
6. a) What are the APIs to connect database in java? Write the advantages of API. 7
b) Explain about the ResultSet. Write the steps to connect database in java. 8
7. Write short notes on: (Any two) 8
a) Delegation Event Model 2×5
b) Difference between JDK and JRE
c) Virtual Machine Vs runtime machine

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Semester: Fall

Year : 2016
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The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe the features of Procedure, Event and Object programming? 7
 b) What do you mean by Thick and Thin Client? Explain N-tier Architecture. 8
2. a) What is delegate in C#? Illustrate with a meaningful example. 7
 b) How dot net application is created? Explain each process in brief. 8
3. a) How to make database connectivity in asp.net? Explain with example. 5
 b) Explain, why Java is called a platform independent language? 5
 c) Define Exception in Java. How it is handled? Explain with example. 5
4. a) Write a swing application with two input textboxes and a button which when clicked, it should write the result of two string in textboxes to console after converting them to Uppercase. 8
 b) Explain about the lifecycle of Applet? Write a simple but meaning applet program. 7
5. a) Explain HTTP requests handling and Session management in JSP. 8
 b) Explain about the life cycle of JSP with a neat diagram? 7
6. a) Define Server side database. Explain the types of Server side database drives and tools. 8
 b) Write a JSP application that submits the basic student registration form to another page called prcoessRegister.jsp which should display all the records. 7
7. Write short notes on: (Any two) 2×5
 - a) Document View Architecture
 - b) Event handling
 - c) CLR and CTS

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Programming Technology

Semester: Spring

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) How problem is solve using aspect oriented programming? Explain the major objects in document view architecture. 7
b) How model, view and controller layers work in MVC architecture? Explain advantages and disadvantages of 2- tire and 3-tier architecture. 8
2. a) Define Delegate. Differentiate between simple array and Array List with example. 8
b) Discuss about different data types and type Conversions in C# with suitable code snippets. 7
3. a) Define Common Language Runtime, Common Type System, Windows Communication Foundation and garbage collection. 8
b) How does garbage collection work in C#? Explain the phases of garbage collection. 7
4. a) Explain the main features of asp.net , write a sample program to create a table in a database, store any data in NET framework and display the contort in label using ADO.Net Architecture (consider asp.net page) 8
b) Discuss the architecture of Java Virtual Machine and its components using suitable diagram. 7
5. a) What is checked and unchecked Exception? Write a sample program which illustrates the concepts of user defined exception in Java. 8
b) Write a code to create simple applet in Java. How do you convert applet into application? Illustrate with an example. 7
6. a) What is Interface in Java? What are the types of events that MouseListener and MouseMotionListener can generate? Discuss with code snippet. 8
b) What are java servlets? Develop a Simple Servlet application i.e LifeCycle application, which will demonstrate the implementation of the init(), service() and destroy() methods. 7
7. Write short notes on: (Any two) 2×5
 - a) JDBC drivers
 - b) JSP Standard Tag Library
 - c) Virtual machines and runtime environments.

POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Programme: BE
Course: Programming Technology

Year : 2020
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are the basic characteristic features of object-oriented programming? How it is different from event-oriented programming and aspect-oriented programming? 7
b) Discuss the advantages of the Document/View Architecture along with Single Document Interface and Multiple Document Interface 8
2. a) How request and response are handled in MVC? Compare between 2-tier and 3-tier architecture. 7
b) What do you mean by event and delegates? Explain simple array and array list with suitable example. 8
3. a) What is .net framework? Explain the relationship between CLS, CTS, CLR and base class library in .net framework. 7
b) Describe the steps involved in establishing database connectivity in asp.net with code snippets? 8
4. a) Explain briefly about JVM, JRE and JDK with their block diagram. 8
b) How error is different from exception? Write a sample program which illustrates the use of try, catch and finally in Java. 7
5. a) What are advantages and disadvantages of Applet? Explain life cycle of an Applet. 7
b) How you handle the event in Swing? Write a java swing program with a button to display "BE" as button name and "PROGRAMMING TECHNOLOGY" as frame name. 8
6. a) What are the advantages of JSP? Write a java program to demonstrate the concepts of reading parameter in Java servlet. 8
b) Define JSP? What are the session management techniques? Explain any two techniques. 7

7. Write short notes on: (Any two)

- a) JDBC vs ODBC
- b) Types of statements in JDBC
- c) Versioning in Dot Net

2×5